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1) $1 \leq q < 2$

$$X = 2^5 + 2^{-10} + 2^{-19}$$

$$X = 2^5 (1 + 2^{-15} + 2^{-24})$$

$$X = (1.000 \dots \dots \underset{\substack{\downarrow \\ a_{15}}}{1} \dots \dots \underset{\substack{\downarrow \\ a_{24}}}{1})_2 \times 2^5$$

$$X_- = (1.00 \dots \dots \underset{\substack{\downarrow \\ a_{15}}}{1} \dots \dots)_2 + 2^{-5} = 2^5 (1 + 2^{-15})$$

$$X_+ = (1.00 \dots \dots \underset{\substack{\downarrow \\ a_{15}}}{1} \dots \dots \underset{\substack{\downarrow \\ a_{23}}}{1})_2 \times 2^5 (1 + 2^{-15} + 2^{-23})$$

4) $f(x) = x^3 + 2x^2 - 5x - 6$

$$X_1 = 1 - \frac{1 + 2 - 5 - 6}{3 + 4 - 5} = 5$$

$$X_0 = 1$$

$$f'(x) = 3x^2 + 4x - 5$$

$$X_2 = 5 - \frac{5^3 + 2 \cdot 5^2 - 5 \cdot 5 - 6}{3 \cdot 5^2 + 4 \cdot 5 - 5} = 3,4$$

$$X = X_0 - \frac{f(X_0)}{f'(X_0)}$$

3) $f(x) = x^3 + 2x^2 - 5x - 6$

$$X_0 = 1 \quad X_1 = -1$$

$$X_{n+1} = X_n - f(X_n) \frac{X_n - X_{n-1}}{f(X_n) - f(X_{n-1})}$$

$$f(X_0) = -8 \quad f(X_1) = 0$$

$$X_2 = -1 - 0 \times \frac{-1 - 1}{0 - (-8)} = -1$$

2) $X = 0.643646995$

$y = 0.643641737$

a) $x + y = 1.28734863 \Rightarrow \text{real}$

$x + y = 1.297349 \Rightarrow \text{measured}$

Absolute = $|\text{real} - \text{measured}| = 0,00000037$

relative = $\left| \frac{\text{Absolute}}{\text{real}} \right| \times 100 = 0,0000289\%$

b) $x - y = 0,000005$

Error = $0,0000289\%$